1.4 Exercises

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1

4

2

If a and c are poth perfect squares, we can check to see if $b = 2mn$ for $a = m^2,\ c = n^2$. If this is the case, we can use the perfect square trinomial pattern. If we only have two terms a and c, we could then possibly use the difference of squares pattern.

3

$2x^2 + 5x + 3$
$\rightarrow (2x + 3)(x + 1)$

4

$3n^2 + 7n + 4$
$\rightarrow (3n + 4)(n + 1)$

5

$4r^2 + 5r + 1$
$\rightarrow (4r + 1)(r + 1)$

6

$6p^2 + 5p + 1$
$\rightarrow (3p + 1)(2p + 1)$

7

$11z^2 + 2z - 9$
$\rightarrow (11z - 9)(z + 1)$

8

$15x^2 - 2x - 8$
$\rightarrow (5x - 4)(3x + 2)$

9

Cannot be factored.

10

$14m^2 + m - 3$
$\rightarrow (7m - 3)(2m + 1)$

11

$9d^2 - 13d - 10$
$\rightarrow (9d + 5)(d - 2)$

12

$5x^2 + 14x - 3$
$\rightarrow (5x - 1)(x + 3)$

13

$9x^2 - 1$
$\rightarrow (3x)^2 - 1$
$\rightarrow (3x + 1)(3x - 1)$

14

$4r^2 - 25$
$\rightarrow (2r)^2 - 25$
$\rightarrow (2r + 5)(2r - 5)$

15

$49n^2 - 16$
$\rightarrow (7n)^2 - 16$
$\rightarrow (7n + 4)(7n - 4)$

16

$16s^2 + 8s + 1$
$\rightarrow (4s)^2 + 8s + 1$

$\rightarrow (4s+1)^2$

17

$49x^2 + 70x + 25$
$\rightarrow (7x)^2 + 70x + 5^2$
$\rightarrow (7x+5)^2$

18

$64w^2 + 144w + 81$
$\rightarrow (8w)^2 + 144w + 9^2$
$\rightarrow (8w+9)^2$

19

$9p^2 - 12p + 4$
$\rightarrow (3p)^2 - 12p + 2^2$
$\rightarrow (3p-2)^2$

20

$25t^2 - 30t + 9$
$\rightarrow (5t)^2 - 30t + 3^2$
$\rightarrow (5t-3)^2$

21

$36x^2 - 84x + 49$
$\rightarrow (6x)^2 - 84x + 7^2$
$\rightarrow (6x-7)^2$

22

$12x^2 - 4x - 40$
$\rightarrow 4(3x^2 - x - 10)$
$\rightarrow 4((3x + 5)(x - 2))$

23

$18z^2 + 36z + 16$
$\rightarrow 2(9z^2 + 18z + 8)$
$\rightarrow 2((3z + 2)(3z + 4))$

24

$32v^2 - 2$
$\rightarrow 2(16v^2 - 1)$
$\rightarrow 2((4v + 1)(4v - 1))$

25

$6u^2 - 24u$
$\rightarrow 6u(u - 4)$

26

$12m^2 - 36m + 27$
$\rightarrow 3(4m^2 - 12m + 9)$
$\rightarrow 3((2m)^2 - 12m + 3^2)$
$\rightarrow 3((2m - 3)^2)$

27

$20x^2 + 124x + 24$
$\rightarrow 4(5x^2 + 31x + 6)$
$\rightarrow 4((5x + 1)(x + 6))$

28

$21x^2 - 77x - 28$
$\rightarrow 7(3x^2 - 11x - 4)$
$\rightarrow 7((3x + 1)(x - 4))$

29

$-36n^2 + 48n - 15$
$\rightarrow -3(12n^2 - 16n + 5)$
$\rightarrow -3((6n - 5)(2n - 1))$

30

$-8y^2 + 28y - 60$
$\rightarrow -4(2y^2 - 7y + 15)$

31

When the monomial was factored out, it was not taken out from the term -36. Instead, it should have been:
$4x^2 - 36$
$\rightarrow 4(x^2 - 9)$
$\rightarrow 4((x + 3)(x - 3))$

32

$16x^2 - 1 = 0$
$\rightarrow (4x + 1)(4x - 1) = 0$
$\rightarrow x = \frac{1}{4} \vee x = -\frac{1}{4}$

33

$11q^2 - 44 = 0$
$\rightarrow 11(q^2 - 4) = 0$
$\rightarrow 11(q + 2)(q - 2) = 0$
$\rightarrow q = 2 \vee q = -2$

34

$14s^2 - 21s = 0$
$\rightarrow 7s(2s - 3) = 0$
$\rightarrow s = \frac{3}{2}$

35

$45n^2 + 10n = 0$
$\rightarrow 5n(9n + 2) = 0$
$\rightarrow n = -\frac{2}{9}$

36

$4x^2 - 20x + 25 = 0$
$\rightarrow (2x)^2 - 20x + 5^2 = 0$
$\rightarrow (2x - 5)^2 = 0$
$\rightarrow x = \frac{5}{2}$

37

$4p^2 + 12p + 9 = 0$
$\rightarrow (2p)^2 + 12p + 3^2 = 0$
$\rightarrow (2p + 3)^2 = 0$
$\rightarrow p = -\frac{3}{2}$

38

$15x^2 + 7x - 2 = 0$
$\rightarrow (5x - 1)(3x + 2) = 0$
$\rightarrow x = \frac{1}{5} \vee x = -\frac{2}{3}$

39

$6r^2 - 7r - 5 = 0$
$\rightarrow (2r + 1)(3r - 5) = 0$
$\rightarrow x = -\frac{1}{2} \vee x = \frac{5}{3}$

40

$36z^2 + 96z + 15 = 0$
$\rightarrow 3(12z^2 + 32z + 5) = 0$
$\rightarrow 3(6z + 1)(2z + 5) = 0$
$\rightarrow z = -\frac{1}{6} \vee z = -\frac{5}{2}$

41

$4x^2 - 19x - 5 = 0$
$\rightarrow (4x + 1)(x - 5) = 0$
$\rightarrow x = -\frac{1}{4} \vee x = 5$

42

$3x^2 - 8x + 5 = 0$
$\rightarrow (3x - 5)(x - 1) = 0$
$\rightarrow x = \frac{5}{3} \vee x = 1$

43

$5x^2 - 27x - 18 = 0$
$\rightarrow (5x + 3)(x - 6) = 0$
$\rightarrow x = -\frac{3}{5} \vee x = 6$

44

$3x^2 - 3x = 0$
$\rightarrow 3x(x - 1) = 0$
$\rightarrow (3x - 0)(x - 1) = 0$
$\rightarrow x = 0 \vee x = 1$

45

$11x^2 - 19x - 6 = 0$
$\rightarrow (11x + 3)(x - 2) = 0$
$\rightarrow x = -\frac{3}{11} \vee x = 2$

46

$16x^2 - 2x - 5 = 0$
$\rightarrow (8x - 5)(2x + 1) = 0$
$\rightarrow x = \frac{5}{8} \vee x = -\frac{1}{2}$

47

$15x^2 - 5x - 20 = 0$
$\rightarrow (15x - 20)(x + 1) = 0$
$\rightarrow x = \frac{20}{15} \vee x = -1$

48

$18x^2 - 6x - 4 = 0$
$\rightarrow (6x - 4)(3x + 1) = 0$
$\rightarrow x = \frac{4}{6} \vee x = -\frac{1}{3}$

49

$12x^2 + 5x - 7 = 0$
$\rightarrow (12x - 7)(x + 1) = 0$
$\rightarrow x = \frac{7}{12} \vee x = -1$

50

$(2x)^2 = 36$
$\rightarrow (2x)^2 - 36 = 0$
$\rightarrow (2x + 6)(2x - 6) = 0$
$\rightarrow x = -3 \vee x = 3$

51

$(3x + 1)x = 30$
$\rightarrow 3x^2 + x - 30 = 0$
$\rightarrow (3x + 10)(x - 3) = 0$
$\rightarrow x = -\frac{10}{3} \vee x = 3$

52

$\frac{1}{2}2x(5x-2) = 115$
$\rightarrow 5x^2 - 2x - 115 = 0$
$\rightarrow (5x+23)(x-5) = 0$
$\rightarrow x = -\frac{23}{5} \vee x = 5$, and thus, because only positive values are accepted, $x = 5$.

53

$2x^2 - 4x - 8 = -x^2 + x$
$\rightarrow 3x^2 - 5x - 8 = 0$
$\rightarrow (3x-8)(x+1) = 0$
$\rightarrow x = \frac{8}{3} \vee x = -1$

54

$24x^2 + 8x + 2 = 5 - 6x$
$\rightarrow 24x^2 + 14x - 3 = 0$
$\rightarrow (6x-1)(4x+3) = 0$
$\rightarrow x = \frac{1}{6} \vee x = -\frac{3}{4}$

55

$18x^2 - 22x = 28$
$\rightarrow 18x^2 - 22x - 28 = 0$
$\rightarrow (9x+7)(2x-4) = 0$
$\rightarrow x = -\frac{7}{9} \vee x = 2$

56

$13x^2 + 21x = -5x^2 + 22$
$\rightarrow 18x^2 + 21x - 22 = 0$
$\rightarrow (6x+11)(3x-2) = 0$
$\rightarrow x = -\frac{11}{6} \vee x = 23$

57

$x = 4x^2 - 15x$
$\rightarrow 4x^2 - 16x = 0$
$\rightarrow (4x - 0)(x - 4) = 0$
$\rightarrow x = 0 \vee x = 4$

58

$(x + 8)^2 = 16 - x^2 + 9x$
$\rightarrow x^2 + 16x + 64 = 16 - x^2 + 9x$
$\rightarrow 2x^2 + 7x + 48 = 0$
$\rightarrow$, because this cannot be factored, the equation has no solution.

59

$2x^3 - 5x^2 + 3x$
$\rightarrow x(2x^2 - 5x + 3)$
$\rightarrow x(2x + 1)(x - 3)$

60

$8x^4 - 8x^3 - 6x^2$
$\rightarrow 2x^2(4x^2 - 4x - 3)$
$\rightarrow 2x^2(2x + 1)(2x - 3)$

61

$9x^3 - 4x$
$\rightarrow x(9x^2 - 4)$
$\rightarrow x(3x + 2)(3x - 2)$

62

$4 = (2 + 2x)(1 + 2x) - 2$
$\rightarrow 4 = 4x^2 + 6x$
$\rightarrow 4x^2 + 6x - 4$
$\rightarrow 2(2x^2 + 3x - 2) = 0$
$\rightarrow 2(2x - 1)(x + 2) = 0$
$\rightarrow x = \frac{1}{2} \vee x = -2$, and thus, because only positive values are accepted, $x = \frac{1}{2}$

63

$96 = (12 + 2x)(8 + 2x) - 96$
$\rightarrow 4x^2 + 40x - 96 = 0$
$\rightarrow (4x + 48)(x - 2) = 0$
$\rightarrow x = -48 \vee x = 2$ Because only positive values are accepted, the width of the border should be 2 ft.

64

$R(x) = (500 - 20x)(45 + 5x)$
$\rightarrow -100(x - 25)(x + 9)$. Thus, the zeros of the function are $25, -9$, meaning the vertex lies on the line $x = -\frac{25-9}{2} = -8$, and thus, the store should sell each board for $-20(-8) = 160$ more, and, therefore, D is the correct answer.

65

First, we set up a quadratic equation to model this situation:
$R(x) = (330 + 15x) \times (6 - 0.25x)$
Next, we find the vertex of this equation, by finding it's zeros:
$-3.75(x + 22)(x - 24)$, and thus, the zeros are -22 and 24, and thus the x-coordinate of the vertex is $\frac{-22+24}{2} = 1$, and thus, the maximum value is $(330 + 15(1))(6-0.25(1)) = \$1983.75$ when each sandwich is sold for $6-0.25(1) = \$5.75$

66

$714 = (25 + 4x)(21 + 2x) - 525$
$\rightarrow 8x^2 + 134x - 714 = 0$
$\rightarrow (4x - 17)(2x + 42) = 0$
$\rightarrow x = \frac{17}{4} \vee x = -21$, and thus, because only positive values are accepted, the width of the mat is 42 ft., while the width at the top and bottom of the paintings is 29.5 ft.

67

(a): The girth is $108 - 36 = 72$ ft.
(b): For the width: $72 = 2w + 2h$
$\rightarrow w = 36 - h$
For the volume: $V = wlh = 36(36 - h)h$
(c): We can rewrite the above formula as a quadratic formula, in order to find the x-coordiante of its vertex:
$V = -36h^2 + 1296h$
$\rightarrow xcord = -\frac{1296}{2(-36)} = 18$, and thus, the maximum volume is $-36(18)^2 + 1296(18) = 11664$. This is the maximum volume, as it is the maximum height of the quadratic function. So, a height of 18 in. and a width of $36 - 18 = 18$ in. maximize the volume of the box, at $11664in^3$.

68

$2x(5x - 4) = (3x + 2)(x + 1)$
$\rightarrow 10x^2 - 8x = 3x^2 + 5x + 2$
$\rightarrow 7x^2 - 13x - 2 = 0$
$\rightarrow (7x + 1)(x - 2) = 0$
$\rightarrow x = -\frac{1}{7} \vee x = 2$, and thus, because only positive values are accpeted, $x = 2$.